

CHOLESTEROL



Cholesterol: Understanding Your Lipid Panel

Cholesterol isn't inherently bad; your body actually needs it. Understanding how cholesterol is transported in your blood can tell you a lot about cardiovascular health.

Cholesterol is a waxy, fat-like molecule found throughout the body. It is essential for normal biological functions, including cell structure, hormone production, vitamin D synthesis, and digestion. Your liver produces the cholesterol your body needs. Because cholesterol doesn't travel freely through blood, it is transported inside particles called lipoproteins. This is where LDL and HDL come in.

Marker	What it tells you
Total Cholesterol	Overall amount of cholesterol measured in your blood
LDL	Carries cholesterol through the body
HDL	Helps transport cholesterol back to the liver for removal
Triglycerides	A form of fat circulating in the blood that the body can use for energy

What can contribute to unhealthy cholesterol levels?

Factors include diet high in saturated/trans fats, low physical activity, smoking, obesity, type 2 diabetes, age, and genetics/family history. Some risk factors are modifiable while others are not.

Always discuss your results with your healthcare provider for proper interpretation.



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LDL - the "bad" cholesterol

LDL stands for low-density lipoprotein and is responsible for transporting cholesterol from the liver to cells throughout the body. Cholesterol itself is essential for building cell membranes and producing certain hormones, but too much LDL cholesterol in the bloodstream can become problematic.

Excess LDL can enter and accumulate within artery walls, where it contributes to inflammation and the development of atherosclerotic plaques.

HDL - the "good" cholesterol

HDL stands for high-density lipoprotein. One of its important functions is reverse cholesterol transport, helping move excess cholesterol away from peripheral tissues and toward the liver.

The liver can then process cholesterol and ultimately eliminate some of it from the body through bile. This cholesterol-transport role is one reason HDL has traditionally been called "good cholesterol."

Biomarker	General optimal level
Total cholesterol	About 150 mg/dL
LDL	About 100 mg/dL
HDL – men	≥40 mg/dL
HDL – women	≥50 mg/dL
Triglycerides	<150 mg/dL



Scientific Facts

Cholesterol is not the enemy:

Every cell needs cholesterol, and it is used to produce hormones, vitamin D, and substances involved in digestion.

High cholesterol can be completely silent:

It usually doesn't cause symptoms, which is why a blood test is needed to know your levels.

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